

CHAPTER 5: WIRELESS NETWORKING

A **wireless network** is a computer network that uses wireless data connections between network nodes. **Wireless networking** allows homes, telecommunications networks, and business installations to avoid the costly process of introducing cables into a building, or as a connection between various equipment locations.

WIRELESS STANDARDS AND TECHNOLOGIES:

IEEE 802.11 STANDARDS

The Evolutionary Timeline

- **802.11b (The Pioneer - 1999)**
 - This was the first widely adopted Wi-Fi. It was slow by today's standards, but it changed the world by removing the ethernet cable.
 - **Speed:** 11 Mbps
 - **Frequency:** 2.4 GHz
- **802.11g (The Multi-Tasker - 2003)**
 - This version kept the long range of "b" but increased the speed significantly.
 - **Speed:** 54 Mbps
 - **Frequency:** 2.4 GHz
- **802.11n (Wi-Fi 4 - 2009)**
 - This was a massive jump because it introduced **MIMO** (Multiple Input, Multiple Output). This means the router could use multiple antennas to send data at once.
 - **Speed:** Up to 600 Mbps
 - **Frequency:** 2.4 GHz and 5 GHz (Dual Band)

The Modern Era

- **802.11ac (Wi-Fi 5 - 2013)**
 - This version focused almost entirely on the **5 GHz** band to avoid the "noise" and interference of the 2.4 GHz band.
 - **Speed:** 1.3 Gbps (1300 Mbps) or higher
 - **Frequency:** 5 GHz only (usually paired with 802.11n for 2.4 GHz support)
- **802.11ax (Wi-Fi 6 / 6E - 2019)**
 - This is the current "Gold Standard." It isn't just about raw speed; it's about **efficiency**. It uses a technology called **OFDMA** which allows the router to talk to 30 devices at the exact same time without them waiting in line.
 - **Speed:** 9.6 Gbps
 - **Frequency:** 2.4 GHz, 5 GHz, and **6 GHz** (Wi-Fi 6E)

Why do the names change? (Wi-Fi 4, 5, 6)

The IEEE names (802.11ax, etc.) were too confusing for regular people. In 2018, the **Wi-Fi Alliance** simplified them:

- **802.11n -> Wi-Fi 4**
- **802.11ac -> Wi-Fi 5**

- **802.11ax -> Wi-Fi 6**

Standard	Frequency	Max Speed	Best Uses
802.11b/g	2.4 GHz	11/54 Mbps	Legacy Devices
802.11n (Wi-Fi 4)	2.4 / 5 GHz	600 Mbps	Basic Home Browsing
802.11ac (Wi-Fi 5)	5 GHz	1.3 Gbps	Gaming and 4K Streaming
802.11ax (Wi-Fi 6)	2.4 / 5 / 6 GHz	9.6 Gbps	Crowded areas (Malls/Schools)

FREQUENCY BANDS

Frequency Bands are the specific ranges of radio waves used to transmit data.

1. The 2.4 GHz Band

This was the first band used for Wi-Fi. It consists of 11 channels (in the US/Philippines) that are 20 MHz wide.

- **Pros:** It has a **long range** and can penetrate solid objects like concrete walls, heavy furniture, and floors very effectively.
- **Cons:** It is very **crowded**. Not only do neighbors' Wi-Fi use it, but so do baby monitors, Bluetooth devices, and microwaves.

2. The 5 GHz Band

As internet speeds increased, the 2.4 GHz band became too slow. The 5 GHz band was introduced to provide more space and speed.

- **Pros:** Much **higher speeds** and much less interference. It has 24 non-overlapping channels, meaning 24 routers can sit next to each other without touching each other's signals.
- **Cons:** It has a **short range**. It struggles to go through walls. If you move two rooms away from the router, your 5 GHz signal might drop significantly.

3. The 6 GHz Band

Introduced with **Wi-Fi 6E**, this is the newest frontier in wireless tech. It opens up a massive amount of new "spectrum."

- **Pros:** Zero interference from older devices. It is extremely fast and has very low "latency" (lag), which is perfect for Virtual Reality (VR) and high-end gaming.
- **Cons:** Requires brand new hardware (Wi-Fi 6E or Wi-Fi 7). Its range is even shorter than 5 GHz.

WIRELESS NETWORK CONFIGURATION:

An **SSID (Service Set Identifier)** is simply the technical name for a Wi-Fi network. When you open the Wi-Fi settings on your phone and see a list of available connections like "Campus_Guest" or "Home_WiFi," you are looking at a list of SSIDs.

It acts as a **unique identifier** so that wireless devices can distinguish one network from another in a crowded area.

1. How the SSID Works

The Wireless Access Point (WAP) or Router acts like a lighthouse. It constantly sends out a small radio packet called a **Beacon Frame** every 100 milliseconds. This beacon contains the SSID name, telling all nearby devices, "I am here, and my name is X."

- **Max Length:** An SSID can be up to **32 characters** long.
- **Case Sensitivity:** CampusWiFi and campuswifi are seen as two completely different networks.
- **Special Characters:** You can use spaces and symbols (like SGU-Staff_5G!), though some older devices might struggle to connect to them.

2. SSID Broadcasting vs. Hidden SSIDs

By default, routers **broadcast** their SSID so anyone can find them. However, you can turn this off in the settings.

- **Broadcasting Enabled:** The network is visible. Users just click and enter a password.
- **Broadcasting Disabled (Hidden SSID):** The network name is not sent in the beacon. To connect, a user must manually type the **exact name** and the **exact password**.

3. Multiple SSIDs on One Device

Modern enterprise routers can broadcast multiple SSIDs at the same time using a single physical antenna. This is called **BSSID (Basic Service Set Identifier)** mapping.

- **Example:** A single router in the Library can broadcast:
 1. SGU_Library (VLAN 10 - For students)
 2. SGU_Faculty (VLAN 20 - For teachers)
 3. SGU_IOT (VLAN 30 - For smart lights and sensors)

4. SSID Roaming

If you have a large building with 10 different Access Points, you should give them all the **exact same SSID** and security settings.

- **The Result:** As a student walks from the North Wing to the South Wing, their phone will automatically "hand off" to the stronger signal without ever disconnecting. This is called **Seamless Roaming**.

INFRASTRUCTURE VS. AD-HOC MODE

In wireless networking, **Mode** refers to how the "conversation" is organized between devices. Think of it like the difference between a **structured classroom** and a **casual group chat**.

1. Infrastructure Mode

This is the most common type of wireless network. In this mode, all devices (Laptops, Phones, Tablets) must communicate through a central "hub" called a **Wireless Access Point (WAP)** or a Wireless Router.

- **The "BSS" (Basic Service Set):** The Access Point and all its connected clients form a "Service Set."

Advantages:

- **Scalability:** You can add 50+ students easily.
 - **Centralized Security:** You set one password on the router for everyone.
 - **Internet Access:** The router usually has a "Gate" (WAN port) leading to the ISP.
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2. Ad-hoc Mode

Ad-hoc (Latin for "for this purpose") is a decentralized, temporary network. There is **no router** and **no access point**. Devices talk directly to each other, "Peer-to-Peer."

- **How it Works:** One device "starts" the network, and others join it. Since there is no central manager, the devices must handle their own coordination.
- **The "IBSS" (Independent Basic Service Set):** It is independent because it doesn't need a wired backbone or a router to function.

Advantages:

- **Instant Setup:** No hardware needed except the devices themselves.
- **Cheap:** No need to buy a \$100\$ router for a one-time file transfer.
- **Portable:** Works in the middle of a forest or an airplane.

SECURITY CONSIDERATIONS IN WIRELESS NETWORKS:

WPA2/WPA3 ENCRYPTION

In the world of wireless networking, **Encryption** is the process of scrambling your data into a secret code. Without it, anyone with a simple antenna could "sniff" your Wi-Fi signals out of the air and read your passwords, bank details, or private chats.

1. WPA2 (Wi-Fi Protected Access 2)

Released in 2004, WPA2 has been the global standard for nearly two decades. It uses a very strong encryption algorithm called **AES (Advanced Encryption Standard)**.

- **How it works (The 4-Way Handshake):** When you type your password, the router and your phone perform a 4-step "handshake" to prove they both know the secret without actually sending the password over the air.

- **The Weakness:** It is vulnerable to **Offline Dictionary Attacks**. A hacker can capture that "handshake" data, take it home, and use a powerful computer to try millions of passwords per second until they find a match.

2. WPA3 (Wi-Fi Protected Access 3)

Released in 2018, WPA3 is the "Next Generation" of security. It was designed to fix the holes found in WPA2.

- **How it works (SAE - Simultaneous Authentication of Equals):** This replaces the old handshake. It is much more secure because it creates a unique encryption key for every single session.
- **Protection against "Brute Force":** Even if you have a weak password like password123, WPA3 makes it almost impossible for a hacker to use an automated "dictionary" to guess it. The router will simply stop responding to repeated wrong guesses.
- **Forward Secrecy:** If a hacker captures your encrypted data today and somehow steals your Wi-Fi password next year, they *still* cannot go back and decrypt the old data they captured.

SSID BROADCASTING & MAC FILTERING

In the world of Wireless Security, **SSID Broadcasting** and **MAC Filtering** are often called "First Line of Defense" measures. While they aren't as strong as WPA3 encryption, they add extra layers of privacy to your campus or home network.

1. SSID Broadcasting (The "Hidden" Network)

By default, a Wireless Access Point (WAP) acts like a billboard, constantly shouting its name so devices can find it. This is called **Broadcasting**.

- **How it Works:** The router sends out "Beacon Frames" every few milliseconds. These frames contain the SSID (e.g., SGU_Admin).
- **Disabling Broadcast:** When you "Hide" the SSID, the router still works, but it removes its name from the beacon. To a normal smartphone, the area looks like it has no Wi-Fi.
- **The Connection Process:** To connect to a hidden network, you must manually go to "Add Network" on your phone and type the **exact name** (case-sensitive) and password.

2. MAC Filtering (The "Guest List")

Every network card in the world (in your phone, laptop, or printer) has a permanent, unique hardware ID called a **MAC Address** (e.g., 00:0A:95:9D:68:16).

MAC Filtering is like a bouncer at a club who has a physical list of names. Even if you have the "ticket" (the Wi-Fi password), if your "name" (MAC Address) isn't on the list, you aren't getting in.

- **White-listing (Allow):** You list only the specific devices you own. Anyone else is blocked automatically.
- **Black-listing (Deny):** You block one specific device (like a neighbor who stole your password) while letting everyone else in.

References:

- Cisco Networking Academy. Networking Concepts and Routing & Switching Modules. Cisco Systems, Inc.